26900374

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026 - Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMBG303 –

EMERGING AND RE-EMERGING DISEASES

(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I), and Appreciation(Ap)

Part A - Very Short Answer Questions - Answer all questions - Each question carries 1 mark

1. **Explain the importance of early diagnosis in disease management. [U] / [CO2]**
   Helps in early treatment, prevents complications and reduces disease transmission.
2. **What is the primary mode of transmission of bubonic plague? [K] / [CO1]**
   Bite of infected rat fleas (Xenopsylla cheopis).
3. **What type of vector transmits Lyme disease? [K] / [CO3]**
   Ticks (Ixodes species).
4. **What is meant by human demographics? [K] / [CO2]**
   Study of human population characteristics such as age, sex, size and distribution.
5. **Name the genus of fungus responsible for mucormycosis. [K] / [CO4]**
   Rhizopus/ Rhizomucor/ Mucor.

[1x5 = 5]

Part B - Short Answer Questions - Answer any 10 questions - Each question carries 2 marks

6. **Mention two factors contributing to coronavirus emergence. [K] / [CO1]**

   Animal-to-human transmission (zoonotic). Increased human–animal contact and international travel. **Any 2 [2]**

7. **Explain how microbial adaptation can contribute to the emergence of infectious diseases. [U] / [CO2]**

   Microbes may undergo genetic mutations and adaptations that help them survive, spread, or infect new hosts, leading to the emergence of infectious diseases.

8. **Write any two symptoms of Influenza. [U] / [CO3]**

   Fever, Cough [2]

9. Define acute Chagas disease. [K] / [CO4]

Acute Chagas disease is the early stage of infection caused by Trypanosoma cruzi [1], usually characterised by fever, swelling at the site of infection and body aches [1].

10. Write any two vaccines for COVID. [U] / [CO3]

Covishield, Covaxin or any other [2]

11. What was Robert Koch's contribution to cholera? [K] / [CO1]

Robert Koch identified *Vibrio cholerae* as the causative agent of cholera in 1883. [2]

12. Why is quarantine used during some infectious disease outbreaks? [U] / [CO2]

Quarantine separates people who may have been exposed to an infectious disease to prevent its spread. [2]

13. Why is typhus called a war disease? [K] / [CO1]

Typhus spreads easily in crowded and unhygienic conditions, especially among soldiers during wars. [2]

14. What are the treatments recommended for emergomycosis? [K] / [CO4]

Antifungal drugs such as amphotericin B and itraconazole are used to treat emergomycosis.[2]

15. Explain the relationship between antimicrobial use and the development of resistance.[U] / [CO2]

Improper or excessive use of antimicrobials creates selection pressure, allowing resistant microbes to survive and multiply.[2]

16. Differentiate between EIDs and REIDs. [U] / [CO1]

• EIDs (Emerging Infectious Diseases): Diseases that are newly appearing or increasing in incidence or geographical spread.[1]

• REIDs (Re-emerging Infectious Diseases): Previously controlled diseases that are increasing again.[1]

17. State two social factors that influence the spread of infectious diseases. [K] / [CO2]

Overcrowding [1], Poor sanitation and hygiene[1]

[2x10 = 20]

Part C - Short Essay Questions - Answer any 5 questions. Each question carries 6 marks

18. Give an account of Yersinia pestis infection, including its epidemiology, mode of transmission and prevention. [U] / [CO3]

- Yersinia pestis is a Gram-negative bacterium that causes plague. [1], found in rodents and other small mammals. [1], - It spreads mainly through the bite of infected rat fleas

(Xenopsylla cheopis). [1]. - It also spreads through contact with infected animals and respiratory droplets in pneumonic plague. [1]. - Symptoms include fever, chills and swollen lymph nodes. [1]. - Prevention includes rodent and flea control, sanitation and early antibiotic treatment. [1]

19. Explain the amastigote and promastigote forms of Leishmania. [U] / [CO4]

Amastigote: Round or oval-shaped, non-flagellated, found inside human macrophages.[3]

Promastigote: Elongated in shape; has a single anterior flagellum.- Found in the gut of sandflies and transmitted to humans.[3]

20. Explain how environmental factors influence the occurrence and transmission of disease. [U] / [CO2]

Climate change affects the survival of disease-causing organisms and vectors. Deforestation increases contact between humans and wildlife. Water pollution spreads waterborne diseases. Poor sanitation increases infectious disease transmission. Urbanisation and overcrowding favour disease spread. Floods and droughts increase the risk of disease outbreaks.[6]

21. Discuss preparedness and response strategies for bioterrorism. [U] / [CO1]

Preparedness: Establish disease surveillance systems.[1], Train healthcare workers [1], maintain vaccines and medicines [1]. Strengthen laboratory facilities [1]. Response: Rapid identification and reporting of cases [1]. Isolation, treatment and public health coordination [1].

22. Explain the role of laboratory surveillance in controlling EIDs. [U] / [CO2]

Helps in early detection of emerging diseases; identifies disease-causing microorganisms. Confirms suspected infections. Monitors disease outbreaks.Detects antimicrobial resistance. Provides information for disease prevention and control.[6]

23. Explain the role of fruit bats in the transmission of Nipah virus. [U] / [CO3]

Fruit bats are natural reservoirs of Nipah virus. They shed the virus through saliva and urine. Contaminated fruits and date palm sap can transmit the virus. Humans may become infected by consuming contaminated food. Infection can spread from animals such as pigs to humans. Human-to-human transmission occurs through close contact with infected persons.[6]

24. Describe the classification of EIDs and REIDs based on mode of transmission. [U] / [CO1]

Airborne: Tuberculosis, influenza. Droplet: Influenza, COVID-19. **Direct contact**: Ebola. **Food- and water-borne**: Cholera, typhoid. **Vector-borne**: Dengue, plague. **Zoonotic**: Nipah, rabies.[6]

[6x5 = 30]

Part D - Essay Questions - Answer any 1 question. Each question carries 15 marks

25. Write an essay on Mycobacterium ulcerans infection. [K] / [CO3]

Introduction and causative organism – [2], Mode of transmission – [2], - Epidemiology and risk factors – [2], - Pathogenesis and role of mycolactone toxin – [3], - Clinical features – [3], - Diagnosis – [1.5], - Treatment and prevention – [1.5]

26. Explain the public health importance of yellow fever and measures adopted for its control. [U] / [CO1]

Introduction and causative agent – [2], Epidemiology and geographical distribution – [2], Mode of transmission and mosquito vector – [2], Clinical features and complications – [3], Public health importance – [2], Prevention through vaccination and mosquito control – [2], Surveillance and outbreak control – [2]

[15x1 = 15]